

Fall 1988

Problem 1 (Fa88, Fa10). Let R be a finite ring. Prove that there are positive integers m and n with $m > n$ such that $x^m = x^n$ for every x in R .

Problem 2 (Fa88, Sp97). Determine the group $\text{Aut}_{\mathbb{C}}$ of all one-to-one analytic maps of $\mathbb{C}$ onto $\mathbb{C}$.

Problem 3 (Fa88). Let the real valued functions $f_1, \dots, f_{n+1}$ on $\mathbb{R}$ satisfy the system of differential equations

$$\begin{aligned}f'_{k+1} + f'_k &= (k+1)f_{k+1} - kf_k & k &= 1, \dots, n \\f'_{n+1} &= -(n+1)f_{n+1}\end{aligned}$$

Prove that for each k ,

$$\lim_{t \rightarrow \infty} f_k(t) = 0$$

Problem 4 (Fa88). Find the Jordan Canonical Form of the matrix

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}$$

Problem 5 (Fa88, Fa08, Sp22). Let f be a continuous, strictly increasing function from $[0, \infty)$ onto $[0, \infty)$ and let $g = f^{-1}$. Prove that

$$\int_0^a f(x) \, dx + \int_0^b g(y) \, dy \geq ab$$

for all positive numbers a and b , and determine the condition for equality. Use this to prove Young's Inequality, which states that if p and q are positive reals with $1/p + 1/q = 1$ and a and b are positive reals then

$$\frac{a^p}{p} + \frac{b^q}{q} \geq ab$$

Problem 6 (Fa88, Sp97). Let f be a function from $[0, 1]$ into itself whose graph

$$G_f = \{(x, f(x)) \mid x \in [0, 1]\}$$

is a closed subset of the unit square. Prove that f is continuous.

Note: See also ????

Problem 7 (Fa88). Find all abelian groups of order 8, up to isomorphism. Then identify which type occurs in each of

1. $\mathbb{Z}_{15}^*$
2. $\mathbb{Z}_{17}^*/\{\pm 1\}$
3. the roots of $z^8 - 1$ in $\mathbb{C}$
4. $\mathbb{F}_8^+$
5. $\mathbb{Z}_{16}^*$

where $\mathbb{F}_8$ is the field of eight elements, $\mathbb{F}_8^+$ is its underlying additive group, and R^* is the group of invertible elements in the ring R , under multiplication.

Problem 8 (Fa88). Do the functions $f(z) = e^z + z$ and $g(z) = ze^z + 1$ have the same number of zeros in the strip $-\pi/2 < \Im z < \pi/2$?

Problem 9 (Sp85, Fa88). Let A and B be two $n \times n$ self-adjoint, i.e., Hermitian, matrices over $\mathbb{C}$ such that all eigenvalues of A lie in $[a, a']$ and all eigenvalues of B lie in $[b, b']$. Show that all eigenvalues of $A + B$ lie in $[a + b, a' + b']$.

Problem 10 (Fa88). Find, up to isomorphism, all groups of order $2p$, where p is a prime.

Problem 11 (Fa88). Let f be an analytic function on a disc D whose center is the point z_0 . Assume that $|f'(z) - f'(z_0)| < |f'(z_0)|$ on D . Prove that f is one-to-one on D .

Problem 12 (Fa88). Let n be a positive integer and let f be a polynomial in $\mathbb{R}[x]$ of degree n . Prove that there are real numbers $a_0, a_1, \dots, a_n$, not all equal to zero, such that the polynomial

$$\sum_{i=0}^n a_i x^{2^i}$$

is divisible by f .

Problem 13 (Fa88). Let A be a complex $n \times n$ matrix, and let $C(A)$ be the commutant of A . Prove that $\dim C(A) \geq n$.

Problem 14 (Fa88, Fa97, Fa24). Let the group G be generated by two elements, a and b , both of order 2. Prove that G has a subgroup of index 2.

Problem 15 (Fa88, Fa09). If f is a smooth real valued function of x, y, z such that

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} > 0$$

for all x, y, z , show that f does not have a local maximum.

Problem 16 (Fa88). Let n be a positive integer. Prove that the polynomial

$$f(x) = \sum_{i=0}^n \frac{x^i}{i!} = 1 + x + \frac{x^2}{2} + \cdots + \frac{x^n}{n!}$$

in $\mathbb{R}[x]$ has n distinct complex zeros, $z_1, z_2, \dots, z_n$, and that they satisfy

$$\sum_{i=1}^n z_i^{-j} = 0 \quad \text{for } 2 \leq j \leq n$$

Problem 17 (Fa88). Prove that $\int_0^\infty \frac{x}{e^x - e^{-x}} dx = \frac{\pi^2}{8}$.

Problem 18 (Fa88). Let g be a continuous real valued function on $[0, 1]$. Prove that there exists a continuous real valued function f on $[0, 1]$ satisfying the equation

$$f(x) - \int_0^x f(x-t)e^{-t^2} dt = g(x)$$